Phase 5 - fairness evidence

Goal: Make the fairness claim measurable with repeatable test traffic, a labeled baseline, and a buyer-readable outcome. Estimated time: 4:45-5:30. Start only after the buyer flow and Phase 3 concurrency checks pass.

Build

1. Add a Node.js/TypeScript simulator using the same HTTP API as buyer clients. Implement ordinary, same-account retry, and flood profiles. Cap concurrency and label every run and client. Optionally add a multi-account profile as a known identity limitation.
2. Store each run's seed, eligible population, inventory, request schedule, client profiles, start/end times, and results in simulation_runs. Isolate runs from the live demo sale and from each other.
3. Run a simple first-request-wins baseline on the same logical population, inventory, and seed. Label it as a model, not a real competitor system. Keep its outcomes separate from live orders.
4. Build /admin/events/[eventId]/lab with simulation controls and a results table. Show allocation rate by cohort, request-to-allocation ratio, throttled requests, oversells, duplicate orders, and queue integrity violations. Show p50/p95 latency only if collected consistently. Charts are optional.
5. Add a buyer fairness receipt showing accepted entry time, ordering policy/version, final position, admission and purchase outcome, and reason codes. Do not imply that a queue position guarantees a ticket.
6. Show configuration and limitations next to every result: simulated logical buyers are not concurrent browser connections; one account per person is not proven; account farms and production-scale load are untested.

Acceptance checks

- Re-running the same configuration produces comparable outcomes without mixing run IDs or corrupting demo orders.
- Same-account retries do not create additional eligible entries or confirmed orders.
- The Lab clearly separates baseline model results from Fair Drop API results and displays request counts alongside allocations.
- A buyer can understand their own outcome from the receipt and reason code.

Handoff to Phase 6

Phase 6 needs a reproducible simulator command and the exact claim supported by the measurements. Record sample configuration, seed, result location, measured scale, latency method if any, and limitations.

Handoff record

Fill in after implementation: files/commits, commands, results, and open issues.